

Mid 461

20 Marks

Q1:

1. Define Computer Graphics:
2. Difference between Absolute and Relative Coordinate Input :
3. A screen has a horizontal length of 10 inches and a vertical height of 5 inches. What is the aspect ratio given that the screen resolution is 1920 x 1080 points?

Q2: True or False

1. Plasma displays are more energy-efficient than LCD displays : *I'm unsure about the Q*
2. $M + Y + C = \text{White}$: False
3. Raster scan device scans the screen from vertical to horizontal : False
4. Raster scan is more efficient than random scan in drawing lines and shapes. : False
5. The refresh rate is the measure of how many times the image on a screen is updated per second. : (True)
6. Graphic applications are used only for entertainment (False)

Q3: Multiple Choice

1. Cathode Ray Tube monitors (CRT)
2. Subtractive Mixing (CMYK)
3. The frequency at which a picture is redrawn on the screen is referred to as the (refresh rate).
4. Computer Graphics Goals : (Synthetic images indistinguishable from reality)
5. Ray emitted by cathode ray tube are : (electrons)
6. Application of computer graphics : (all of the above)
7. Digital display use : (RGB)

Q4: Fill in the Blanks

1. Yellow + Magenta = Red
2. Magenta + Cyan = Blue
3. Green + Blue = Cyan
4. (resolution) The maximum number of points/pixels that can be displayed (without overlap) on a CRT is referred to as the.
5. (Persistence) is defined as the time that it takes the emitted light from the screen to decay to one-tenth of its original intensity.
6. (Color gamut) The set of all colors that can be produced from mixing the primary colors.
7. (Tint) White pigment added to pure pigment.
8. (Tone) The consequence of adding both white and black pigments to pure pigments.

Q5: List four of

1. Hard-copy devices :
 - Laser printer
 - Inkjet printer
 - Film recorder
 - Electrostatic printer
 - Pen plotter
 2. Applications of Computer Graphics:
 - Video games
 - Simulation and virtual reality
 - Computer-aided design (CAD)
 - Medical imaging
 - Education and training
 - Film production and animation
- Q : RGB stands for : Red , Green , Blue

Q6: Using Bresenham's Algorithm

Starting at (3 , 3)

Ending (10 , 7)